Degradation under Gaussian weight perturbation: muon / adamw

chinchilla (token budget)
model size
30M
60M
100M
300M
600M
128
64
32
16
8
4
2
1
0.5
0.25
γ = 0.003
2.64x
3.85x
3.09x 2.59x 2.99x
2.23x 2.12x 3.89x
2.13x 1.94x 2.67x 1.26x
1.44x 2.36x 1.35x 1.23x
1.41x 2.06x 1.48x 1.81x 1.37x
2.38x 1.91x 1.67x 1.72x 1.08x
2.49x 1.31x 0.78x 1.31x 1.09x
0.59x 3.59x 0.64x 1.29x
γ = 0.005
2.67x
3.87x
3.13x 2.52x 3.08x
2.43x 2.07x 3.51x
2.60x 2.12x 2.84x 1.34x
1.84x 2.43x 1.78x 1.22x
1.82x 2.02x 1.70x 1.38x 1.58x
2.29x 1.49x 1.72x 1.35x 1.21x
3.88x 1.73x 1.20x 1.24x 1.23x
1.35x 2.25x 2.10x 1.08x 1.08x
γ = 0.007
2.83x
3.77x
3.00x 2.43x 3.27x
2.32x 2.14x 3.42x
2.51x 2.11x 2.82x 1.25x
1.67x 2.38x 1.71x 1.09x
1.75x 1.90x 1.77x 1.38x 1.48x
2.41x 1.60x 1.68x 1.30x 1.26x
2.85x 1.73x 1.33x 1.22x 1.23x
0.72x 2.56x 2.34x 1.24x 1.22x
γ = 0.01
2.90x
3.55x
2.96x 2.46x 3.30x
2.31x 2.25x 3.27x
2.53x 2.12x 2.77x 1.16x
1.76x 2.52x 1.72x 1.15x
1.83x 1.83x 1.85x 1.45x 1.48x
2.30x 1.61x 1.70x 1.33x 1.31x
2.46x 1.60x 1.39x 1.29x 1.29x
1.46x 2.52x 1.95x 1.26x 1.21x
γ = 0.02
2.99x
3.77x
2.98x 2.45x 3.47x
2.37x 2.29x 3.22x
2.51x 2.16x 2.75x 1.15x
1.80x 2.45x 1.78x 1.19x
1.86x 1.80x 1.85x 1.39x 1.49x
2.45x 1.64x 1.68x 1.29x 1.28x
3.11x 1.77x 1.49x 1.29x 1.26x
1.91x 2.15x 1.89x 1.23x 1.18x
γ = 0.03
3.10x
3.74x
2.99x 2.50x 3.59x
2.35x 2.33x 3.35x
2.52x 2.17x 2.76x 1.11x
1.84x 2.50x 1.80x 1.18x
1.89x 1.81x 1.90x 1.37x 1.47x
2.44x 1.64x 1.67x 1.27x 1.30x
3.23x 1.80x 1.50x 1.30x 1.27x
2.21x 1.97x 1.87x 1.23x 1.18x
γ = 0.04
3.20x
3.79x
2.95x 2.52x 3.69x
2.39x 2.37x 3.40x
2.53x 2.21x 2.85x 1.11x
1.85x 2.50x 1.82x 1.22x
1.93x 1.80x 1.91x 1.37x 1.49x
2.44x 1.66x 1.69x 1.27x 1.34x
3.30x 1.84x 1.52x 1.29x 1.30x
2.41x 1.92x 1.87x 1.23x 1.21x
γ = 0.05
3.31x
3.84x
3.00x 2.56x 3.78x
2.40x 2.40x 3.47x
2.51x 2.23x 2.86x 1.11x
1.88x 2.52x 1.83x 1.22x
1.94x 1.81x 1.94x 1.38x 1.49x
2.46x 1.67x 1.70x 1.26x 1.34x
3.37x 1.85x 1.53x 1.32x 1.29x
2.56x 1.87x 1.86x 1.22x 1.21x
γ = 0.07
3.55x
3.84x
2.98x 2.63x 4.05x
2.40x 2.48x 3.67x
2.50x 2.30x 2.98x 1.11x
1.92x 2.57x 1.88x 1.26x
1.94x 1.81x 1.98x 1.39x 1.53x
2.47x 1.70x 1.73x 1.26x 1.37x
3.39x 1.89x 1.55x 1.33x 1.31x
2.68x 1.85x 1.88x 1.23x 1.23x
γ = 0.1
3.71x
3.61x
2.94x 2.71x 4.44x
2.37x 2.59x 4.00x
2.47x 2.39x 3.17x 1.11x
1.96x 2.67x 1.98x 1.37x
1.98x 1.84x 2.06x 1.41x 1.57x
2.46x 1.80x 1.80x 1.27x 1.46x
3.43x 1.97x 1.58x 1.35x 1.35x
2.80x 1.84x 1.93x 1.24x 1.23x
γ = 0.13
3.40x
3.22x
2.82x 2.73x 4.70x
2.21x 2.65x 4.21x
2.40x 2.41x 3.42x 1.08x
2.00x 2.75x 2.07x 1.56x
2.01x 1.86x 2.17x 1.44x 1.58x
2.47x 1.91x 1.90x 1.29x 1.56x
3.44x 2.07x 1.63x 1.39x 1.39x
2.96x 1.86x 1.99x 1.26x 1.25x
2x
1x
0.5x
muon / adamw loss degradation (log scale)

cell = (perturbed - unperturbed) for muon DIVIDED BY the same for adamw | >1x (red) = muon degrades more | white = no data or undefined

colour scale clipped at 0.26x-3.79x (97th pct); 12 cell(s) beyond it keep their printed value